

Numerical Methods in Soft Matter Exercises

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Integration schemes

9.1 First order integrators for the harmonic oscillator

Source code `src/091_first_order.c`

Analysis script `src/091_first_order.R`

Executable `src/091_first_order`

The Hamilton's equations of motion for the one-dimensional harmonic oscillator are:

$$\begin{cases} \dot{p} = \frac{\partial \mathcal{H}}{\partial x} = -kx, \\ \dot{x} = \frac{\partial \mathcal{H}}{\partial p} = \frac{p}{m}, \end{cases} \quad (9.1.1)$$

where the Hamiltonian is given by

$$\mathcal{H} = \frac{p^2}{2m} + \frac{kx^2}{2}. \quad (9.1.2)$$

The exercise asked to compare two integrators, the non-symplectic Euler integrator:

$$\begin{cases} p_{t+\Delta t} = p_t - x_t \Delta t, \\ x_{t+\Delta t} = x_t + p_t \Delta t, \end{cases} \quad (9.1.3)$$

and another first order integrator, but symplectic:

$$\begin{cases} p_{t+\Delta t} = p_t - x_t \Delta t, \\ x_{t+\Delta t} = x_t + p_{t+\Delta t} \Delta t. \end{cases} \quad (9.1.4)$$

If we express eqs. (9.1.3) and (9.1.4) in matrix form, we get

$$\begin{pmatrix} x \\ p \end{pmatrix}_{t+\Delta t} = \begin{pmatrix} 1 & \Delta t \\ -\Delta t & 1 \end{pmatrix} \begin{pmatrix} x \\ p \end{pmatrix}_t, \quad (9.1.5)$$

$$\begin{pmatrix} x \\ p \end{pmatrix}_{t+\Delta t} = \begin{pmatrix} 1 - \Delta t^2 & \Delta t \\ -\Delta t & 1 \end{pmatrix} \begin{pmatrix} x \\ p \end{pmatrix}_t. \quad (9.1.6)$$

The determinants of the update matrices are then

$$\det \begin{pmatrix} 1 & \Delta t \\ -\Delta t & 1 \end{pmatrix} = 1 + \Delta t^2 > 1 \quad (9.1.7)$$

$$\det \begin{pmatrix} 1 - \Delta t^2 & \Delta t \\ -\Delta t & 1 \end{pmatrix} = 1 - \Delta t^2 + \Delta t^2 = 1. \quad (9.1.8)$$

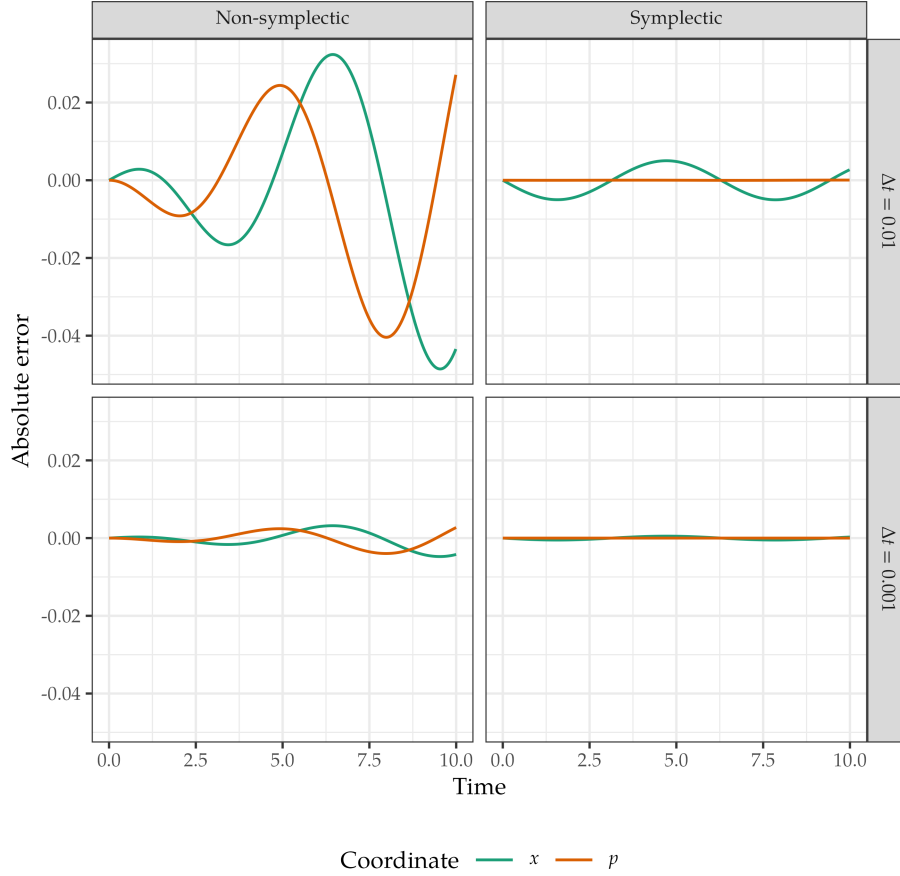


Figure 9.1.1: absolute error of x_t and p_t for the one-dimensional harmonic oscillator integrated with the non-symplectic integrator in eq. (9.1.3) (left panels) and the symplectic integrator in eq. (9.1.4) (right panels). The two upper panels refer to an integration with time step $\Delta t = 0.01$, the bottom two to $\Delta t = 0.001$.

This means that only the integrator in eq. (9.1.4) preserves the phase space measure, while the non-symplectic Euler integrator will display an exponentially growing error in the energy.

Actually, the symplectic integrator in eq. (9.1.4) does not strictly conserve the Hamiltonian $\mathcal{H}$ in eq. (9.1.2), but a *shadow Hamiltonian* $\mathcal{H}'$. If we set for convenience $k = m = 1$, $\mathcal{H}'$ takes the form

$$\mathcal{H}' = \mathcal{H} - \frac{px}{2} \Delta t. \quad (9.1.9)$$

Indeed, if we calculate $\mathcal{H}'$ at time $t + \Delta t$, following the transformation, we get

$$\begin{aligned} \mathcal{H}'(x_{t+\Delta t}, p_{t+\Delta t}) &= \frac{1}{2}(p_t - x_t \Delta t)^2 + \frac{1}{2}[(1 - \Delta t^2)x_t + p_t \Delta t]^2 \\ &\quad - \frac{\Delta t}{2}(p_t - x_t \Delta t)[(1 - \Delta t^2)x_t + p_t \Delta t] \\ &= \frac{1}{2} \left[p_t^2 + \cancel{x_t^2 \Delta t^2} - \cancel{2x_t p_t \Delta t} + x_t^2 + \cancel{x_t^2 \Delta t^4} - \cancel{2x_t^2 \Delta t^2} + \cancel{p_t^2 \Delta t^2} \right. \\ &\quad \left. + \cancel{2x_t p_t \Delta t} - \cancel{2x_t p_t \Delta t^3} - x_t p_t \Delta t + \cancel{x_t p_t \Delta t^3} \right. \\ &\quad \left. - \cancel{p_t^2 \Delta t^2} + \cancel{x_t^2 \Delta t^2} - \cancel{x_t^2 \Delta t^4} + \cancel{x_t p_t \Delta t^3} \right] \\ &= \frac{x_t^2 + p_t^2}{2} - \frac{x_t p_t}{2} \Delta t \\ &= \mathcal{H}(x_t, p_t) - \frac{x_t p_t}{2} \Delta t \\ &= \mathcal{H}'(x_t, p_t). \end{aligned} \quad (9.1.10)$$

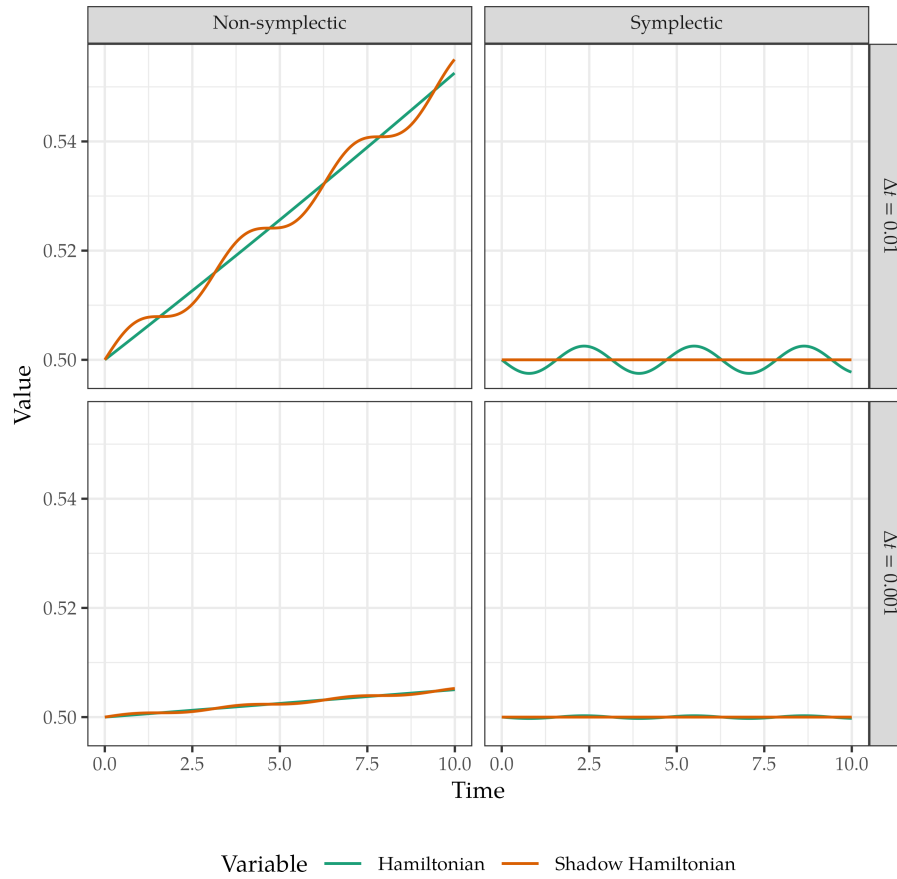


Figure 9.1.2: Hamiltonian and shadow Hamiltonian of the one-dimensional harmonic oscillator integrated with the non-symplectic integrator in eq. (9.1.3) (left panels) and the symplectic integrator in eq. (9.1.4) (right panels). The two upper panels refer to an integration with time step $\Delta t = 0.01$, the bottom two to $\Delta t = 0.001$.

Figure 9.1.1 shows the absolute integration error of the two presented schemes compared to the exact solution $(x_t, p_t) = (\cos t, -\sin t)$ (with starting condition $x_0 = 1, p_0 = 0$) at two different time steps: $\Delta t = 0.01$ and $\Delta t = 0.001$. As expected, the error is smaller with $\Delta t = 0.001$. Additionally, the symplectic integrator's errors remain smaller and stable over time, in contrast to the non-symplectic integrator, whose errors increase with time. Figure 9.1.2 shows the values of the Hamiltonian and the shadow Hamiltonian from eq. (9.1.9) over time. The symplectic integrator correctly preserves $\mathcal{H}'$, while its Hamiltonian displays stationary oscillations around the true value. In the case of the non-symplectic integrator, the Hamiltonian grows linearly in time, and $\mathcal{H}'$ is the one that oscillates around it, also growing in time.

9.2 Higher order integrators

Source code `src/092_higher_order.c`

Executable `exe/092_higher_order`

Inside `src/092_higher_order.c`, I implemented two higher order integration schemes: *velocity Verlet* and *Beeman's algorithm*. In this case, I left the possibility to set k and m different from 1, while the rest of the program is broadly similar to `src/091_first_order.c`. With a generic k , the force at time t is $f(x_t) = -kx_t$, and the velocity Verlet step becomes:

$$p_{t+\Delta t/2} = p_t - \frac{kx_t}{2}\Delta t, \quad (9.2.1)$$

$$x_{t+\Delta t} = x_t + \frac{p_{t+\Delta t/2}}{m}\Delta t, \quad (9.2.2)$$

$$p_{t+\Delta t} = p_{t+\Delta t/2} - \frac{kx_{t+\Delta t}}{2}\Delta t. \quad (9.2.3)$$

Beeman's algorithm provides more accuracy in the momentum, but it needs the force – hence, the position – at the previous time step $t - \Delta t$ to update the system to $t + \Delta t$. The algorithm is implemented as follows:

$$x_{t+\Delta t} = x_t + \frac{p_t}{m}\Delta t - \frac{k(4x_t - x_{t-\Delta t})}{6m}\Delta t^2, \quad (9.2.4)$$

$$p_{t+\Delta t} = p_t + \frac{k}{6}(2x_{t+\Delta t} + 5x_t - x_{t-\Delta t})\Delta t. \quad (9.2.5)$$

Since the position at $t = -\Delta t$ is needed to proceed from the initial condition (x_0, p_0) , I simply used velocity Verlet for the first step of Beeman's.

9.3 Testing the higher order integrators

Source code `src/092_higher_order.c`

Analysis script `src/093_higher_order.R`

Executable `exe/092_higher_order`

For this exercise, I compared the two algorithms presented above in two scenarios: a 'normal' one with $k = m = 1$ and $\Delta t = 0.1$, and a 'problematic' integration with $k = 312.5$, $m = 0.5$ and $\Delta t = 0.1$. The second set of parameters corresponds to $\omega = \sqrt{k/m} = 25$, and since the stability condition for the velocity Verlet algorithm is $\omega\Delta t < 2$, I expected that integrator to fail, and possibly also Beeman's.

Figures 9.3.1 and 9.3.2 display the results in terms of, respectively, the absolute error relative to the exact solution

$$\begin{cases} x_t = \cos(\omega t), \\ p_t = -m\omega \sin(\omega t), \end{cases} \quad (9.3.1)$$

and the relative difference of the Hamiltonian from its initial value, $\mathcal{H}_t/\mathcal{H}_0 - 1$. We can notice immediately that the second scenario, with $\omega\Delta t = 2.5 > 2$, is highly unstable for both

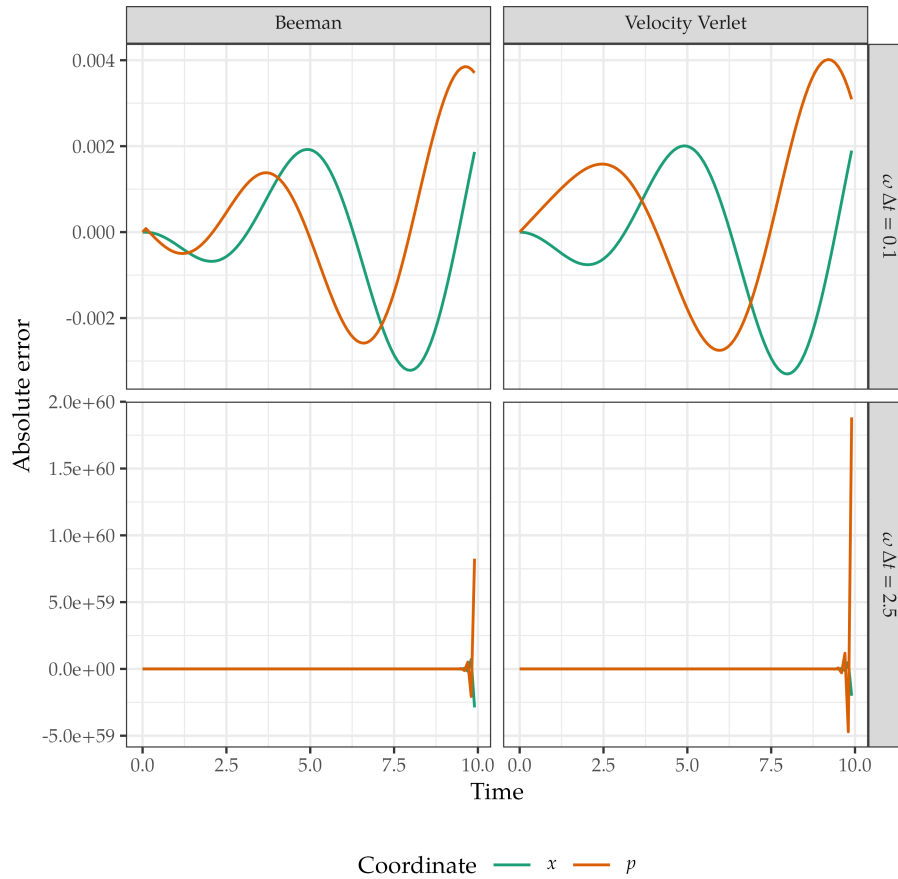


Figure 9.3.1: absolute error of x_t and p_t for the one-dimensional harmonic oscillator integrated with the velocity Verlet scheme (left panels) and Beeman's algorithm (right panels). The two upper panels refer to an integration with $k = m = 1$, the bottom two to one with $k = 312.5$ and $m = 0.5$, while both have $\Delta t = 0.1$.

algorithms: the coordinates, and with them the Hamiltonian, grow exponentially with time. Conversely, in the first scenario both algorithms seem to work properly, with the errors on x and p oscillating and slowly increasing with time, and the Hamiltonian oscillating steadily around a constant value – as expected, since both algorithms are symplectic.

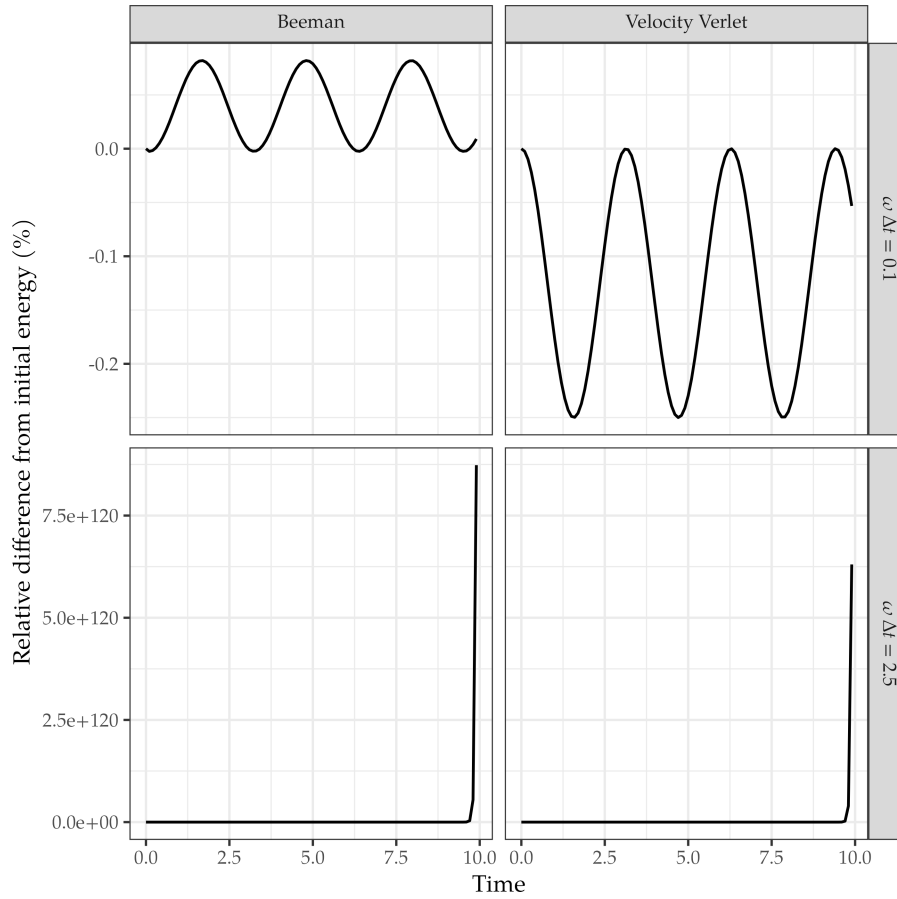


Figure 9.3.2: relative difference of the Hamiltonian from its initial value, $\mathcal{H}_t/\mathcal{H}_0 - 1$, for the one-dimensional harmonic oscillator integrated with the velocity Verlet scheme (left panels) and Beeman's algorithm (right panels). The two upper panels refer to an integration with $k = m = 1$, the bottom two to one with $k = 312.5$ and $m = 0.5$, while both have $\Delta t = 0.1$.